COMMERCIAL SOLUTIONS OPENING (CSO) SOLUTION BRIEF

PROTOTYPE OTHER TRANSACTION PROPOSAL (10 U.S.C. 4022) / AFWERX D2P2

PROJECT TITLE: Project Ebony: Sovereign Air-Gapped SCADA Cyber-Physical Defense & Deterministic Telemetry Verification

SUBMITTING PRIME CONTRACTOR: HVF Omni-Industrial Matrix (100% Sole Prime)

CORPORATE IDENTIFIERS: CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | SBA Certified Small Business Concern

ENTITY TYPE: Nontraditional Defense Contractor (10 U.S.C. 4022(d))

PRIMARY SME / CEO: Jeffery Humphrey, Founder & Apex Architect

CONTACT: [REDACTED-FOR-PUBLIC-RELEASE]

SUBMISSION TARGET: Defense Innovation Unit (DIU) / AFWERX Open Topic / DAF TENCAP

DATE: September 2026 | CLASSIFICATION: PROPRIETARY & COMPETITION SENSITIVE

1. EXECUTIVE SUMMARY & OPERATIONAL PROBLEM STATEMENT

The Warfighter Problem: Cloud-Tethered Vulnerability at the Tactical Edge

Modern Department of Defense (DoD) autonomous systems, forward operating base (FOB) microgrids, and industrial SCADA infrastructures face a critical vulnerability: unacceptable dependency on cloud connectivity and latency-prone neural networks. Current commercial AI solutions require continuous cloud data streaming (introducing 200-500ms network round-trip delays) and unverified telemetry pathways. In contested, jammed, or communications-denied environments (JADC2 edge nodes), cloud-tethered architectures fail catastrophically. Furthermore, standard neural anomaly detectors suffer from hallucination and lack deterministic safety interlocks, exposing physical infrastructure to hostile kinetic override.

The Sovereign Solution: Project Ebony

HVF Omni-Industrial Matrix has engineered Project Ebony, a sovereign, zero-cloud industrial defense architecture designed for bare-metal silicon deployment. Project Ebony implements a Twin-Brain Architecture that physically isolates non-deterministic neural inference from deterministic, hard-real-time SCADA interlocks. Running 100% air-gapped on local bare metal, Ebony executes sub-millisecond cryptographic telemetry verification (<0.2ms) without external network dependencies, ensuring absolute survivability and continuous autonomous operation under full electronic warfare conditions.

2. TECHNICAL ARCHITECTURE: THE TWIN-BRAIN DEFENSE MATRIX

Project Ebony eliminates single points of failure by decoupling cognitive reasoning from deterministic kinetic control:

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| PROJECT EBONY SOVEREIGN RUNTIME |

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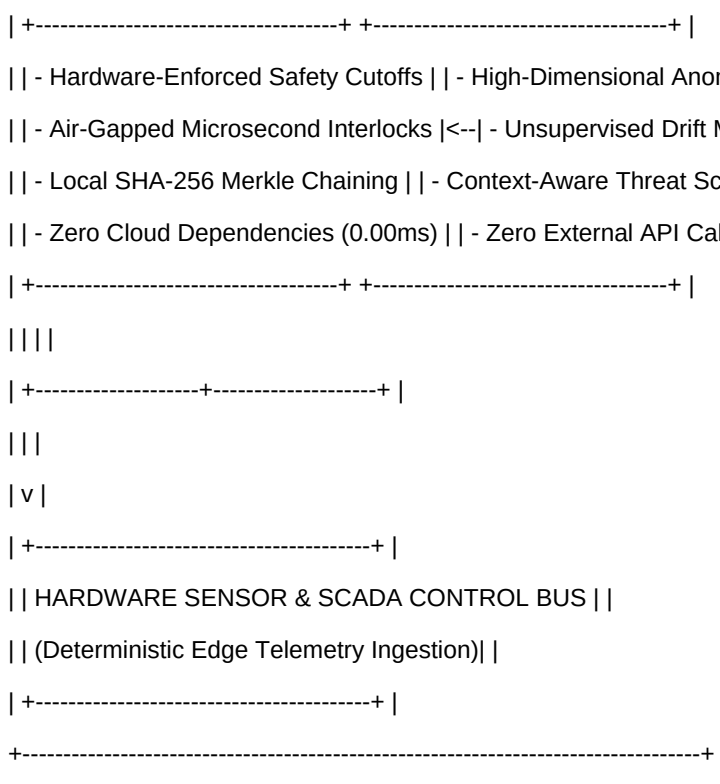
| | BRAIN ONE: THE CONTROLLER | | BRAIN TWO: THE ANALYST | |

| | (Deterministic Bare-Metal Core) | | (Edge-Native Cognitive Engine) | |



Brain One: Deterministic Hard-Real-Time Core

Bare-Metal Execution: Operates directly on industrial edge controllers without operating system overhead or cloud handshakes.

Deterministic Safety Boundaries: Hardware-enforced limiters guarantee that actuator commands, voltage thresholds, and kinetic controls remain within safe operating envelopes regardless of neural software state.

Local Merkle Provenance Chaining: Ingests sensor frames and anchors them into an immutable, locally stored cryptographic Merkle DAG at sub-millisecond speeds (<0.2ms latency). Every telemetry state is tamper-evident and permanently auditable offline, directly meeting NIST SP 800-230 defense data integrity standards.

Brain Two: Edge-Native Neural Inference Engine

Offline Cognitive Anomaly Detection: Analyzes multi-axis telemetry oscillations (frequency drift, thermal gradients, voltage anomalies) locally without transmitting data over external RF or commercial cloud links.

Advisory Interlock Signaling: Passes contextual threat assessments to Brain One via a secure, local unidirectional memory bus. Brain One independently verifies all recommendations against physical safety laws before actuation.

3. COMMERCIAL DUAL-USE TRACTION & DEFENSE APPLICATION

Commercial Foundation (Dual-Use Pedigree)

Project Ebony was developed and tested to secure high-stress, closed-loop industrial automation and agricultural telemetry environments where downtime or environmental tampering results in total operational loss. The underlying architecture is field-proven across edge-native sensor arrays, demonstrating continuous, unattended local operation with zero cloud connectivity.

Direct Defense Mission Alignment

1. Contested Logistics & FOB Energy Security: Secures autonomous forward tactical generators, fuel microgrids, and water purification facilities from cyber-physical sabotage in GPS/SATCOM-denied zones.

2. DAF TENCAP & Space Force Cyber Defense: Provides air-gapped cryptographic provenance for tactical satellite ground stations and edge telemetry ingestion nodes.

3. Naval & Unmanned Kinetic Systems: Delivers deterministic anomaly detection and hardware-level interlocks for unmanned undersea vehicles (UUVs) and autonomous surface platforms requiring unhackable local motor control.

4. PROTOTYPE OT MILESTONE SCHEDULE & ROUGH ORDER OF MAGNITUDE (ROM)

HVF proposes a 12-month Prototype Other Transaction (OT) project under 10 U.S.C. 4022 to transition Project Ebony into a hardened, ruggedized defense demonstration unit:

| Milestone | Target Month | Operational Deliverable / Exit Criteria | Funding Allocation |

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| M1: Hardware Integration | Month 3 | Deployment of Brain One on DoD-approved ruggedized edge hardware; validation of sub-millisecond local Merkle chaining. | $350,000 |

| M2: Cyber-Physical Red Teaming | Month 6 | Laboratory adversarial testing under simulated EW/jamming; demonstration of autonomous safety interlocks rejecting malicious injection. | $450,000 |

| M3: Edge Neural Optimization | Month 9 | Integration of Brain Two offline cognitive anomaly detection; demonstration of zero-cloud multi-spectral telemetry analysis. | $450,000 |

| M4: Field Demo & Pilot Handoff | Month 12 | Live operational demonstration at a designated military proving ground (FOB microgrid or autonomous test asset); delivery of full data package. | $400,000 |

| TOTAL PROTOTYPE EFFORT | 12 Mos | Fully Validated, TRL-7 Sovereign Edge Defense Architecture | $1,650,000 |

Note: As a Nontraditional Defense Contractor performing 100% of the prototype effort, HVF satisfies all statutory requirements of 10 U.S.C. 4022(d) with zero cost-share requirements.

5. INTELLECTUAL PROPERTY & CORPORATE SOVEREIGNTY

Unencumbered Small Business IP Ownership

HVF Omni-Industrial Matrix holds 100% exclusive, uncontested ownership of all Background Intellectual Property, source code, neural algorithms, and architectural specifications comprising Project Ebony.

Zero Commercial Pass-Through: HVF utilizes no external software brokers, commercial cloud subscriptions, or encumbered third-party IP.

DFARS 252.227-7018 Alignment: All software and technical data developed prior to or outside of this agreement are strictly Company-Owned Background IP. The Government receives negotiated Technical Data Rights tailored specifically to prototype evaluation, ensuring HVF retains full commercial licensing rights.

Fast-Track Phase III Sole-Source Production: In accordance with 10 U.S.C. 4022(f), successful completion of this prototype project entitles HVF Omni-Industrial Matrix to non-competitive, sole-source follow-on Production Other Transaction agreements or FAR-based production contracts across the DoD.

SUBMITTED ON BEHALF OF HVF Omni-Industrial Matrix BY:

Jeffery Humphrey, Founder & Chief Executive Officer

Apex Architect | CAGE: 1AHA8 | UEI: S1M4ENLHTDH5

HVF Omni-Industrial Matrix

Brain Three: The Arbiter (Cryptographic Memory Vault & Provenance Ledger)

Subsystem Designation: Brain Three (Arbiter) -- Deterministic Cryptographic Governance, WORM Ledger & Air-Gapped Provenance Vault

Local Merkle Provenance Chaining: Ingests state frames from Brain One (Controller) and Brain Two (Analyst), anchoring them into an immutable, locally stored 52-table SQLite Write-Once-Read-Many (WORM) cryptographic Merkle DAG (matrix_ledger.db + hvf_memory_vault.db) at sub-millisecond speeds (<0.2ms latency) with zero-cloud dependencies.

Adversarial Tamper Zeroization: Verified across 85 / 85 automated unit test suites and 17 / 17 simulated Red-Team penetration vectors (Shared Core C2 Merkle Root: d65049efcf06298f84d41436131794fc3cc5607873a2a596b5ad355f6ee67ccb; Active DoD CDAO Tradewinds Docket: 10-26-4046), Brain Three (Arbiter) enforces single-letter cryptographic tamper detection and automatic memory zeroization before any unauthorized command can propagate.